

M002 Extension 2 — Piston & Logic Maze

Topic: Two Pistons + AND Conditions · **Course:** Maze Madness · **Level:** Extension (Intermediate) ·

Time: about 45 minutes

The agent follows a **redstone trail** through a long maze, using **AND** at some junctions. Twice a **lever powers a piston** to open the way — get the agent through the **whole maze**.

You drive it with chat words: **l** turn left, **r** turn right, **run** start the solver, **r1** teleport back to you.

1 · Read the Chat Commands

Each block binds a **chat word** to an action.

```
def on_chat():
    agent.turn(RIGHT_TURN)
    player.on_chat("r", on_chat)
```

You type **r**. What does the agent do?


```
def on_chat():
    agent.teleport_to_player()
    player.on_chat("r1", on_chat)
```

You type **r1**. Where does the agent go? When does that help in a long maze?

2 · AND at a Junction

AND is true only when **both** parts are true.

```
if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

Fill the table. Write turn or no turn.

Redstone LEFT?	Redstone AHEAD?	What happens?
yes	no	
no	yes	
yes	yes	

What must be true for the agent to turn left?

3 · Trace the Solver 🔍

This loop runs when you type `run`.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)
    elif agent.detect(BLOCK, AHEAD):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

The last `elif` is a piston gate. A block is ahead. What does `agent.interact(AHEAD)` do?

Why can the agent move forward right after it interacts?

4 · Fix the Bug 🐛

The agent should turn left only when redstone is left AND ahead. This turns on either one.

```
if agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

Hint: "left and ahead" — both at once.

Write the fixed line:

Why was it wrong?

5 · Show Your Work 📷 🎥

Open the **M002 EXT 2** world and type **run**. Watch where it sticks, fix it, run again until it reaches the goal past both pistons.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.